

defect types are rare by nature. In many real settings, a factory has fewer than 30 labeled examples of a new inspection task and needs a working classifier within days, not months.

Large vision-language models (VLMs) like GPT-4.1 offer a potential shortcut. These models can interpret images and follow natural language instructions without task-specific training. However, their accuracy on specialized industrial tasks is often poor. On concrete aggregate grading, a 9-class problem defined by DIN 1045 [9], GPT-4.1 achieves only 29.6% accuracy in a few-shot setting. On steel surface defect classification, it reaches 91.1% but requires curated reference images. Performance varies widely across domains, and these models cannot be deployed on-premises due to cost, latency, and data privacy constraints.

Small VLMs in the 1–7B parameter range [10] are attractive for deployment. They fit on edge hardware, run without cloud connectivity, and can be fine-tuned with parameter-efficient methods like LoRA [6]. The problem is that standard fine-tuning on 18–30 images teaches the model to map specific images to labels but does not transfer domain knowledge. The model memorizes rather than learns.

Chain-of-thought (CoT) prompting [11] has shown that including reasoning steps improves model performance on complex tasks. CoT distillation [5] extends this idea by training smaller models on reasoning-augmented data generated by larger models. However, existing CoT distillation work focuses on text-only language models. The SEAL framework [12] demonstrated that models can improve through self-generated training data with answer-conditioned edits, but again only for text tasks.

We propose answer-conditioned CoT distillation for VLMs. The method works as follows: a frontier VLM receives each training image along with its correct label and generates a visual explanation describing why that label is correct, using domain-specific terminology and contrastive reasoning. A small VLM is then fine-tuned on these reasoning-augmented training pairs via LoRA. The frontier model is not classifying. It already knows the answer. It is explaining what visual features justify that answer and why similar classes do not apply.

This design choice is critical. The frontier model’s own classification accuracy ranges from 29.6% to 91.1% across our four tasks. If we asked it to classify and explain (the standard CoT distillation approach), 9% to 70% of the training data would contain incorrect reasoning. By conditioning on the correct answer, we ensure that every training example contains reasoning directed toward the correct conclusion, mitigating the risk of teaching incorrect decision boundaries.

We validate this method on four industrial classification tasks:

- Concrete aggregate grading (9 classes, macro photography) [9]
- Steel surface defect detection (6 classes, grayscale surface images) [13]
- Steel microstructure classification (5 classes, optical/SEM microscopy) [9]
- Weld defect identification (4 classes, X-ray radiography) [14]

Using only 18 to 30 labeled images per task, our method consistently outperforms direct fine-tuning across all four domains. The complete training pipeline (CoT generation + LoRA fine-tuning) completes in under 2 hours per task on a single NVIDIA L4 24GB GPU, making same-day adaptation feasible. On weld radiography, the fine-tuned 3B model outperforms GPT-4.1 few-shot by 10.8 percentage points. On concrete grading, it achieves 79.6% where the base model scores 12.0% and the frontier model scores 29.6%.

Our contributions are: